SUPERMARKET SALES ANALYSIS

EXPLORATORY DATA ANALYSIS AND VISUALIZATION USING PYTHON

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Tools	Python, Google Colab, Pandas, NumPy, Matplotlib, Seaborn

Academic Project Report

Dataset disclosure: The transaction data used in this report is simulated for educational demonstration. It is not collected from an actual supermarket.

Abstract

This project demonstrates Exploratory Data Analysis (EDA) on supermarket transaction records using Python. The workflow includes dataset inspection, duplicate removal, missing-value treatment, descriptive statistics and six visualizations. The analysis explores revenue by product line and branch, payment preferences, transaction-value distribution, quantity relationships and correlations among numerical variables. The dataset is simulated for educational use, so findings illustrate the analysis process and should not be interpreted as real commercial performance.

1. Introduction

Supermarkets generate transaction records containing information about products, quantities, prices, payment methods, branches and customer ratings. EDA helps summarize such data, identify quality issues and reveal patterns that may support business understanding. This project applies a structured EDA workflow using Python libraries in Google Colab.

2. Problem Statement

Raw transaction data can be difficult to interpret without systematic processing. It may contain missing values, duplicate records and multiple variables. This project explores how data cleaning, statistical summaries and visualizations can make transaction information easier to understand.

3. Objectives

1. Inspect the dataset structure and variable types. 2. Identify missing values and duplicate records. 3. Clean the data using appropriate methods. 4. Calculate descriptive statistics. 5. Compare revenue by product line and branch. 6. Examine payment methods and transaction values. 7. Visualize relationships among numerical variables.

4. Dataset Description

The supplied CSV contains 1002 rows before cleaning and 1000 rows after duplicate removal. It includes fields such as Invoice_ID, Branch, City, Customer_type, Gender, Product_line, Unit_price, Quantity, Tax_5_percent, Total, Date, Time, Payment, COGS, Gross_margin_percentage, Gross_income and Rating. A small number of missing values and duplicate rows were intentionally included to demonstrate preprocessing.

5. Tools and Technologies

Python is used as the programming language. Google Colab provides the notebook environment. Pandas supports data loading and manipulation, NumPy supports numerical operations, Matplotlib creates plots and Seaborn supports statistical visualizations.

6. Data Preprocessing

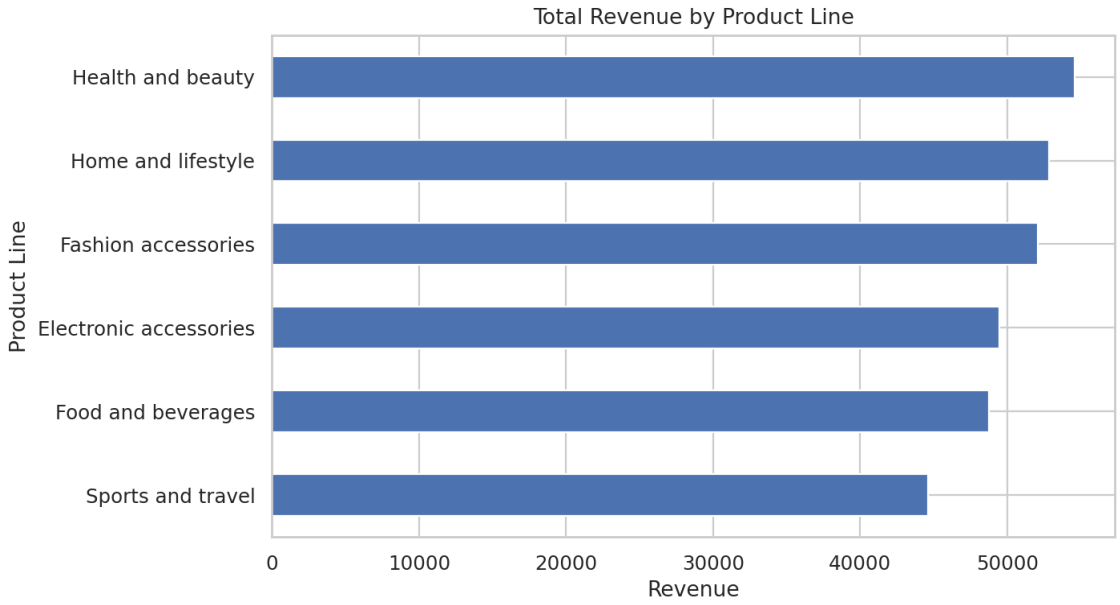
The initial dataset contained 1002 rows. Duplicate removal eliminated 2 repeated rows, leaving 1000 unique records. Missing numerical values were replaced with the median, while missing categorical values were replaced with the mode. The Date column was converted to a datetime type, and derived month and weekday fields were created for further analysis.

7. Exploratory Data Analysis

The cleaned dataset contains 1000 records. The total of the transaction Total column is 302,287.19, and the average transaction value is 302.29. The average customer rating is 7.03 out of 10 in this simulated sample. These descriptive measures summarize the generated dataset only.

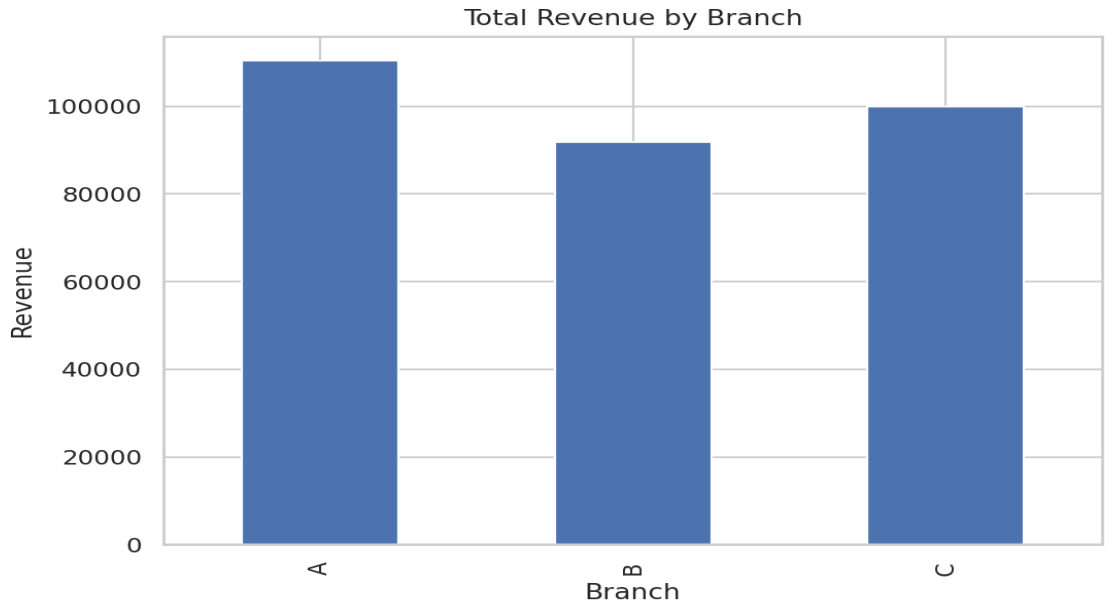
8. Data Visualizations and Observations

Figure 1. Revenue by Product Line



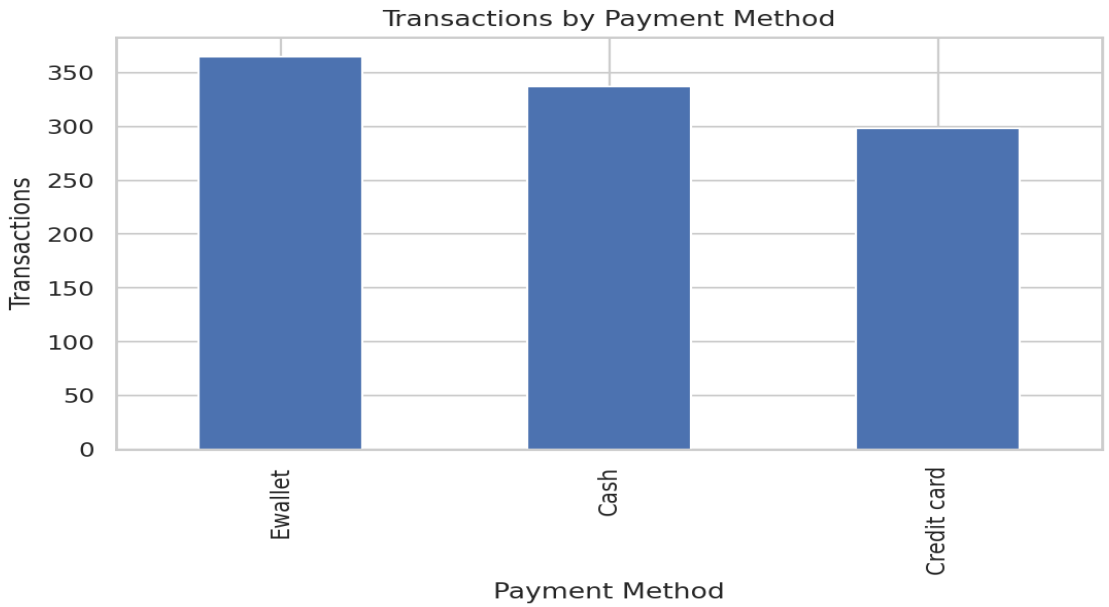
The product line with the largest total revenue in this sample is Health and beauty (54,595.84). The bar chart compares the total revenue across all product lines.

Figure 2. Revenue by Branch



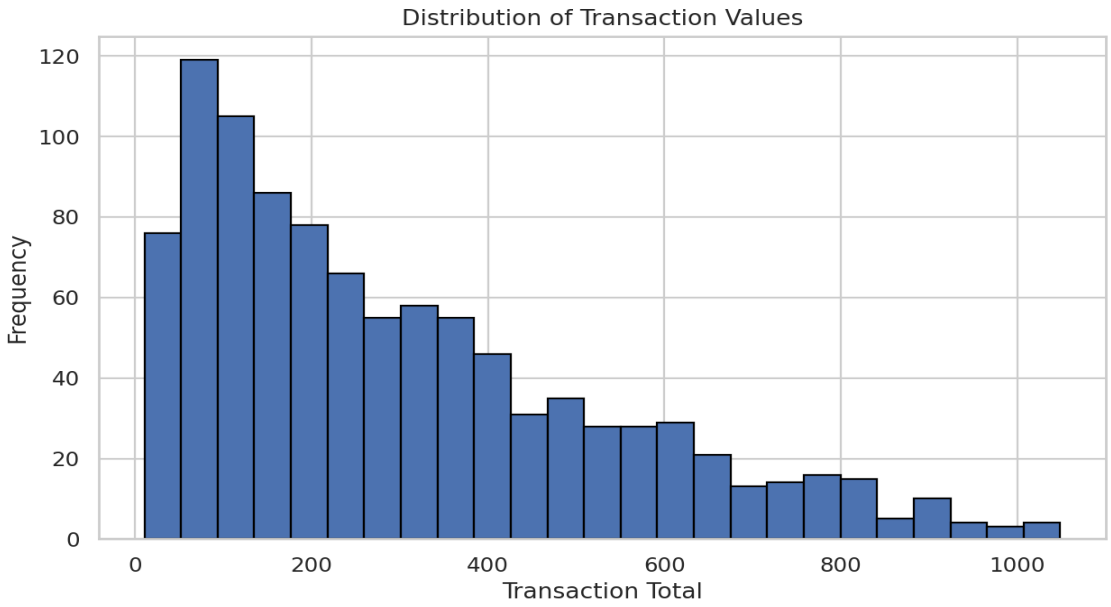
Branch A records the highest total revenue in this sample (110,482.43). The chart compares the transaction totals for branches A, B and C.

Figure 3. Payment Method Distribution



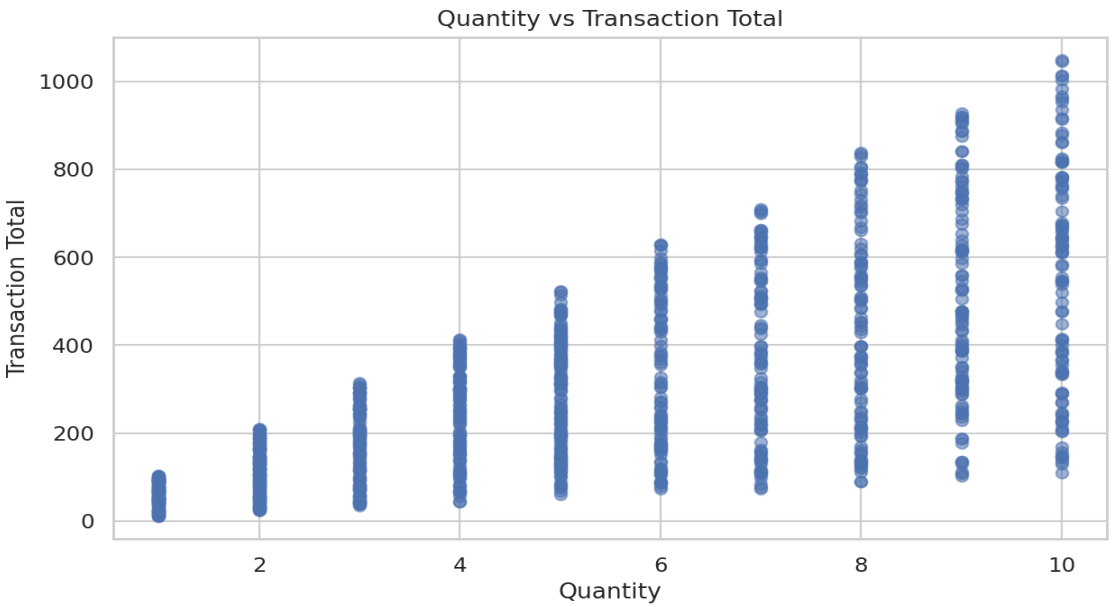
The most frequently recorded payment method is Ewallet with 365 transactions after cleaning. The chart compares the number of transactions for each payment method.

Figure 4. Transaction Value Distribution



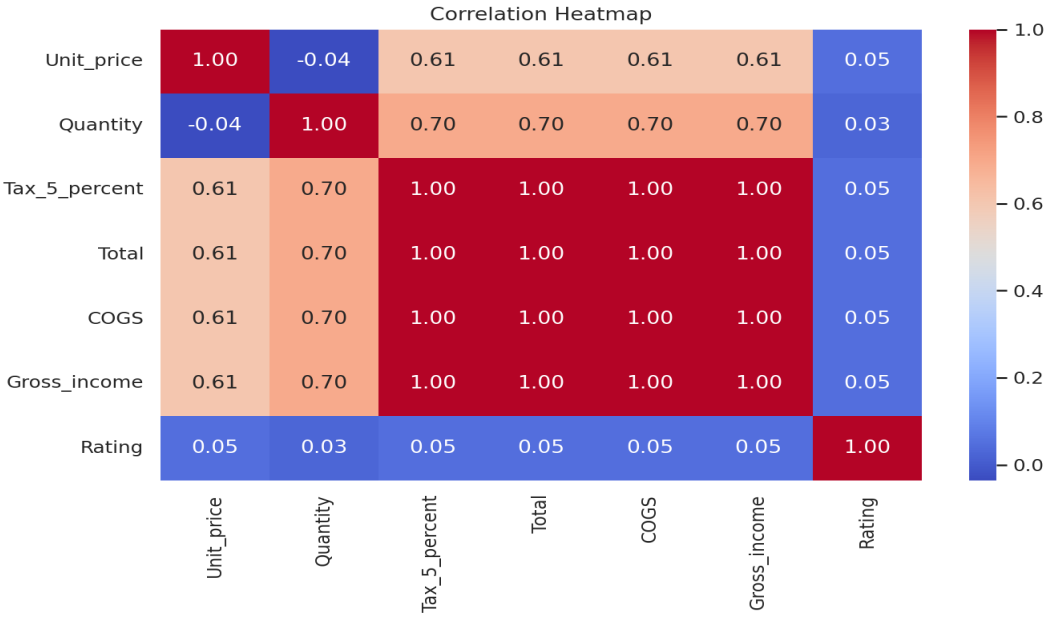
The histogram shows how transaction totals are distributed across the dataset. It helps identify common transaction-value ranges and the spread of spending.

Figure 5. Quantity versus Transaction Total



The scatter plot compares the number of items purchased with the transaction total. It helps inspect how transaction value changes with quantity, while recognizing that unit prices also influence the total.

Figure 6. Correlation Heatmap



The heatmap displays pairwise correlations among selected numerical columns. Values closer to +1 indicate positive linear association, values closer to -1 indicate negative linear association, and values near 0 indicate weak linear association. Correlation does not establish causation.

9. Summary of Findings

• Highest-revenue product line: Health and beauty.
• Highest-revenue branch: A.
• Most frequent payment method: Ewallet.
• Cleaned transaction records: 1000.
• Average transaction value: 302.29.
These findings are calculated from the simulated dataset supplied with the project.

10. Conclusion

The project demonstrates the main stages of an introductory EDA workflow, including data inspection, cleaning, descriptive analysis and visualization. The charts provide a clear way to explore product-line revenue, branch performance, payment preferences, transaction-value distribution, quantity and numerical relationships. The analysis can be repeated with a real, appropriately licensed supermarket dataset to investigate actual business performance.

11. Limitations and Future Enhancement

The main limitation is that the dataset is simulated and does not represent actual supermarket operations. Future work can use a public real-world dataset, perform time-based sales analysis, compare customer segments, create an interactive dashboard and validate findings with domain experts.

References

1. Pandas documentation: https://pandas.pydata.org/docs/
2. NumPy documentation: https://numpy.org/doc/
3. Matplotlib documentation: https://matplotlib.org/stable/
4. Seaborn documentation: https://seaborn.pydata.org/
5. Google Colab: https://colab.research.google.com/